Material safety data sheet

Product name: Flat & Kube

This document is provided as a service for our customers.

Section 1: Product and company identification

Product identification:

Product name: Flat or Kube battery pack
Chemistry: Lithium-ion

Company identification:

Company: TYVA Energie S.A.S.U.
P.A. Marenton 3, 11 rue des Sources
07100 Annonay
FRANCE
Emergency number: + 33 (0)4 26 30 80 79

Section 2: Composition/information on components

Chemical name	No. CAS	Weight /%
75-90 %		
Aluminium	7429-90-5	2-10 %
Copper	7440-50-8	3-15 %
Graphite	7782-42-5	10-30 %
Lithiated cobalt oxide	12190-79-3	20-50 %
Organic Solvents	141-78-6	5-20 %
	96-49-1	EA (Ethyl Acetate)
	616-38-6	EC (Ethylene Carbonate)
	24937-79-9	DMC (Di Methyl Carbonate)
		PVDF Polyvynilidene
		difluoride
10-25 %		
Fr4	65997-17-3	5-15 %
	7440-50-8	
Copper and brass	7440-50-8	1-10 %
Aluminium	7429-90-5	10-20 %
ABS	9003-56-9	<5 %
Polyacetal Copolymer	24969-26-4	<5 %

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Section 3: Hazard(s) identification

The lithium covered in this data sheet is hermetically sealed in an aluminium alloy or metal case and not hazardous if used as recommended by the manufacturer.
The materials contained in this battery may only represent a hazard if the integrity of the battery is compromised or if the battery is physically or electrically abused.

Caution:
Do not open or disassemble.
Do not expose to fire or flame.
Do not mix with batteries of varying sizes, chemistries or types.
Do not puncture, deform incinerate or heat above 60 °C.
Do not short circuit.

Main hazards:
Electrolyte may irritate skin and eyes.

Section 4: First-aid measures

In case of battery rupture of explosion, evacuate personnel from contaminated area and provide maximum ventilation to clear out corrosive fumes/gases and pungent odour.

Eye contact:
Flush with plenty of water during 15 minutes.

Skin contact:
Remove all contaminated clothing and flush affected area with plenty of water.

Inhalation:
Move to fresh air.

Ingestion:
Contact a doctor or a dedicated hospital for poisoning.

Section 5: Firefighting measures

Extinguishing media:
Dry chemical or foam extinguisher; sand.

Special protective equipment:
Use self-contained breathing apparatus to avoid irritant fumes and wear fire protective equipment.

Fire and explosion hazard:
Cells or batteries damaged, opened or exposed to overheating due to external source or due to unproper use can break and leakage can occur.

Special exposure hazards:
Lithium based component, hydrofluoridric acid and carbon oxides.

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Section 6: Accidental release measures

Individual protection:
Use the personal protective equipment. Avoid contact with eyes.

Environmental protection:
Avoid leakage of electrolyte if the cell or battery is opened.

Cleaning process:
Collect the components in specific containers and labelled properly.

Section 7: Handling and storage

Storage:
Store the cell or battery between +5 and +25°C in a ventilated room.
A temperature higher than 60°C could lead to leakage or break.
Humidity might be around 65% (+-10%).
Keep away from heat sources.
Do not allow battery terminals to contact each other or contact with other metals to prevent short-circuit.
Store the cell or battery charged between 20 and 50 % of capacity.

Handling:
Do not crush, pierce, short circuit the terminals with conductive goods.
Do not solder or directly heat.
Do not throw into fire.
Do not mix batteries of different types and brands.
Do not mix new and used batteries.
Keep batteries in a non-conductive tray.
Do not dismantle, damage or abuse the cell or battery mechanically.

Section 8: Exposure controls/personal protection

Personal protection:
Under normal conditions of usage, the batteries don't have any hazard components. No need to check during handling when batteries are not damaged.

Exposure controls:
Under normal conditions of usage, the personnel protective equipment should include safety goggles.

Section 9: Physical and chemical properties

N/A

Section 10: Stability and reactivity

Stability:
The product is stable under normal conditions of use and storage.

Reactivity:
Lithium based component, hydrofluoridric acid and carbon oxides.

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Section 11: Toxicological information

Do not contain toxic materials.

Section 12: Ecological information

No ecological impact is expected under normal conditions of usage.

Section 13: Disposal considerations

Do not throw into fire or submerse the batteries.
Follow the specific requirements for disposal.
Batteries contained recyclable items, so recycling is recommended. Batterie shouldn't be thrown in the nature.

Section 14: Transport information

Classification:
Class 9

UN number:
UN 3480 LITHIUM ION BATTERIES
UN 3481 LITHIUM ION BATTERIES IN/WITH AN EQUIPMENT

Shipping:
Lithium-ion batteries should be conformed with *"UN Manual of test and criteria, Part III, sub-section 38.3".*

Packaging:
The packaging should be certified to the shipping of lithium-ion battery.

Labelling requirements:
Signs and labels should be conformed to the law.

Section 15: Regulatory information

To this battery the following legislation is applicable:

- ADR: road transport
- IMDG: maritime transport
- IATA-DGR: air transport

Section 16: Other information

The information specified in this document are not a guidance regarding the safety during handling, usage, change, storage, transport, disposal and are not a warranty or a technical specification. The details apply only for the designed product and not valid for a use in combination with any other material or in any process.

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